Zecheng Zhou



University of Macau · PhD Student
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Biophysical Modeling | Cardiac Electrophysiology | Epidemiology | Infectious Disease Modeling

Profile

PhD student at the University of Macau working on multi-scale computational modeling of physiological systems. My research integrates ODE-based electrophysiology, calcium-handling dynamics, and biomechanical models to study cardiac arrhythmia mechanisms and organ perfusion.

Previously trained in Preventive Healthcare at Xiamen University, with research experience in infectious disease modeling and chronic disease epidemiology.

Skilled in mechanistic modeling, parameter optimization, and large-scale simulation using MATLAB, Python, and C++. Experienced with statistical modeling approaches including logistic regression, chain binomial models, and PSM-DID.

Education

PhD Student — University of Macau, Macau SAR, China
Sep 2025 - Present

B.Sc. in Preventive Healthcare — Xiamen University, Fujian, China
Sep 2020 - Jun 2025 · GPA: 82.34 / 100 · CET-6: 504 · CET-4: 499

Technical Skills

- **Programming & Analysis:** R, Python, C, MATLAB, web-crawling (Python)
- **Statistical Software:** SPSS, Stata
- **Tools:** EndNote, Microsoft Office, GraphPad Prism
- **Methods:** Logistic regression, chain binomial models, infectious disease modeling, PSM-DID, basic metabolomics analysis

Research Projects

Cardiac Virtual Cell Modeling: Multimodal Data, Biophysical Mechanisms, and AI

PhD Research Project · Aug 2025 - Present
University of Macau

Integrating public single-cell/single-nucleus transcriptomics, fluorescence imaging, and multiscale biophysical models to explore a virtual cell framework combining mechanistic and AI/data-driven approaches for predicting cardiac cell states and dynamic responses.

- **Single-cell/single-nucleus data analysis:** Reanalyzed public human atrial snRNA-seq data using Seurat normalization, Harmony batch correction, cell-subtype annotation, differential expression, and pseudobulk analysis to characterize cell states and disease-associated features.
- **Fluorescence imaging and wet-lab work:** Conducting cell culture, staining, electrical stimulation, and functional assays, including Fluo-4 calcium fluorescence and Di-8-ANEPPS membrane-potential imaging; analyzing ROI signals and fluorescence time series to quantify cellular dynamics.
- **Multiscale mechanistic modeling:** Coupled ODE systems for ion-channel electrophysiology, subcellular calcium regulation, and sarcomere contraction to model intracellular Ca^{2+} dynamics and active force generation.
- **Mechano-electric feedback and arrhythmia mechanisms:** Incorporated stretch-activated currents, mechanical feedback, and spatially heterogeneous calcium buffering; simulated calcium waves and delayed afterdepolarizations (DADs) across pacing frequencies.
- **Parameter analysis and multimodal calibration:** Performed Sobol global sensitivity analysis to quantify parameter effects on calcium transients and arrhythmia triggers; exploring how transcriptomic features and imaging time series can constrain model parameters and support comparison with experimental observations.
- **AI-mechanistic integration (exploratory):** Investigating Neural ODEs, neural-network/ODE hybrid models, and multimodal data fusion for data-driven parameter estimation, cell-state and perturbation-response prediction, and iterative model refinement.

Results:

- Presentation accepted at the 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM-ECCOMAS 2026), Munich, Germany, Jul 2026.

Image-Based Modeling of Liver Perfusion and Metabolism

PhD Research Project · Aug 2025 - Present
University of Macau

- Developing a digital liver perfusion model combining medical imaging, vascular flow modeling, and metabolic simulation.
- Reconstructed 3D liver geometry and vascular structures from CT imaging data.
- Built multi-scale perfusion models describing blood flow through hepatic vascular networks.
- Coupled perfusion with ODE-based metabolic models (oxygen, glucose, lactate transport).
- Investigated temperature-dependent changes in vascular resistance and perfusion dynamics relevant to ex-vivo organ perfusion.
- Developed computational pipelines for simulation of transport and metabolic activity in liver tissue.

Relationship between dietary sialic acid intake and serum HbA1c Project Leader · Jan 2024 - Present

Population-based cross-sectional study, Xiamen, China

- Conducted a cross-sectional analysis of the association between dietary sialic acid intake and HbA1c levels.
- Performed logistic regression and related statistical analyses using SPSS and R.
- Gained experience in nutrient analysis methods and English scientific writing.

Main results:

- Co-authored paper in *Scientific Reports* (JCR Q1): "Relationship between dietary sialic acid intake and serum HbA1c: a population-based study in Xiamen, China."
- Poster presented at the South China Public Health Conference & CHPAMS 2024 Wall Poster Showcase.

Sialic acid intervention, LDL sialylation, and prevention of atherosclerosis Project Leader · Jul 2023 - Present

- Used different substrate combinations to study trans-sialidase (TS) reaction characteristics.
- Explored modification of LDL by exogenous sialic acids and their potential as TS substrates.

Gains: basic laboratory operations and use of laboratory equipment. Results: team won second and third prizes in Xiamen University's Basic Medical Innovative Research and Experimental Design Forum.

Assessment of aggregated HFMD outbreaks in Fujian Province based on a chain binomial model

Project Leader · Jul 2022 - Jun 2023

- Established a chain binomial model based on the natural history of HFMD virus.
- Fitted the model to aggregated outbreak data in Fujian Province to estimate transmission parameters and predict cluster outbreaks.

Gains: strengthened skills in R, Python, infectious disease modeling, experimental planning, and project submission. Results: developed an "Online Geocoding Software Based on R Language" at ShenSoft.

Temporal shifts in 24 notifiable infectious diseases in China before and during COVID-19

Team Member · May 2023

- Crawled provincial-level surveillance data for 24 notifiable diseases.
- Contributed to modeling and prediction for selected diseases using R and Python.

Results:

- Co-authored paper in *Nature Communications* (IF 14.7, JCR Q1).
- Research reported on Xiamen University Science and Technology Department webpages and other media.

Enhancing probiotic effects of *Bifidobacterium bifidum* and preparation of engineered strains

Team Member · Dec 2023 - Present

- Performed metabolomics data analysis and screening of engineered strains.
- Learned project filing, patent drafting, and scientific writing.

Results: awarded Outstanding Project in the 8th Annual Innovation and Entrepreneurship Conference of Xiamen University.

Impact of long-term care insurance pilot on catastrophic medical expenditures in China

Leader · Apr 2024 - Present

- Evaluated the impact of long-term care insurance (LTCI) on catastrophic medical expenditures of urban and rural residents.
- Applied propensity score matching difference-in-differences (PSM-DID) models using Stata.

Results: manuscript under revision for *China Health Statistics* (Peking University Core / Xiamen University First Class Core Journal).

Academic Competitions & Awards

- **National:** College Students' Innovative Entrepreneurial Training Plan Program - National Project.
- **National:** Tenth National Forum for Undergraduate Innovation Research & Experimental Design Competition - Gold Award and Best Manner Award.
- **National:** 14th MathorCup Maths Applied Challenge 2024 - Second Prize (National), Provincial First Prize, Fourth Place in East China Competition.
- **School:** Xiamen University Outstanding Undergraduate Thesis - Second Prize.
- **School:** First and Second National Forum for Undergraduate Innovation Research & Experimental Design Competition, Xiamen University - Second and Third Prizes.
- **School:** 2022 Outstanding Achievements in Practical Teaching of Ideological and Political Theory Courses, Xiamen University - Second Prize.
- **Xiamen University Innovation & Entrepreneurship Conference:** Outstanding Project Award (Bifidobacterium project).

Selected Publications & Presentations

- **Zhou, Z.**, Zhang, W., Wang, X. *et al.* Relationship between dietary sialic acid intake and serum HbA1c: a population-based study in Xiamen, China. *Scientific Reports*, 14, 23026 (2024).
- Li, K., Rui, J., Song, W., Luo, L., Zhao, Y., Qu, H., Liu, H., Wei, H., Zhang, R., Abudunaibi, B., Wang, Y., **Zhou, Z.**, Xiang, T., & Chen, T. Temporal shifts in 24 notifiable infectious diseases in China before and during the COVID-19 pandemic. *Nature Communications*, 15, 3891 (2024).
- Li, K., Wu, J., Zhang, R., Xie, Y., **Zhou, Z.**, Yin, Q., Chen, Q., Rui, J., Guan, X., Zhao, Z., & Chen, T. Global, regional, and national burdens of pertussis among adults: a systematic analysis of age-specific trends using Global Burden of Diseases 2021 data. *Infectious Diseases of Poverty*, 14, 85 (2025).
- **Zhou, Z.**, & Lei, C. Spatial remodeling of myolysis and troponin C buffering couples subcellular Ca^{2+} dynamics to contractile function in atrial fibrillation. Accepted conference presentation, WCCM-ECCOMAS 2026, MS341B: Imaging-Based Mathematical and Computational Modeling in Biomechanics II, Munich, Germany, Jul 2026.

Contact

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